FoodLogAthl-218: Constructing a Real-World Food Image Dataset Using Dietary Management Applications

Taro Yamada

Abstract

We present **FoodLogAthl-218**, a real-world food image dataset constructed from meal photos submitted by users of the FoodLog Athl dietary management application. Unlike existing food image benchmarks sourced from the web, our dataset preserves authentic logging conditions: natural class imbalance, casual photography, multi-dish meal compositions, and temporal continuity across users. The dataset contains 6,925 images, 14,349 dish bounding boxes, and 218 food classes derived from real app records, accompanied by rich metadata including anonymized user identifiers, timestamps, and meal context. We define three benchmark tasks tailored to real-world dietary monitoring: (1) closed-set dish classification on cropped bounding boxes, (2) continual user-aware adaptation along the time axis, and (3) menu/context-aware per-dish classification in multi-dish images. Evaluation with state-of-the-art large multimodal models (LMMs) and CLIP variants reveals that current models struggle significantly with this dataset—GPT-4o achieves only 42.4% top-1 accuracy and an F1 of 0.373. Distribution analysis confirms greater intra-class spread and inter-class overlap compared to the established UEC-Food 256 benchmark. Continual LoRA fine-tuning substantially closes the gap, while user-specific personalization further boosts performance. FoodLogAthl-218 is released publicly to encourage research on realistic, personalized, and context-aware food recognition.

1 Introduction

Automatic food recognition is a prerequisite for scalable dietary logging, nutritional assessment, and personalized health management [3, 26, 28]. Smartphone dietary apps such as FoodLog Athl [28], Asken [4], and MyFitnessPal [9] collect millions of meal photos daily, yet the accuracy of automatic dish identification in this setting remains far below what is required for practical deployment.

A principal obstacle is the *dataset gap*: leading food recognition benchmarks—Food-101 [6], UEC-Food 256 [18], Food2K [27], and Food-500 Cap [24]—are populated predominantly with web-scraped or studio-style images. These images are visually clean, centred, and aesthetically composed, bearing little resemblance to the hurried snapshots that real users capture at mealtimes. Consequently, models trained or evaluated on such data provide an overly optimistic picture of real-world performance.

We address this gap by introducing **FoodLogAthl-218**, a dataset built *bottom-up* from actual FoodLog Athl user records. Rather than defining food classes in advance and sourcing images accordingly, we start from 11,460 meal photos logged between May 2023 and October 2024, apply a reproducible multi-stage curation pipeline, and consolidate the resulting labels into 218 classes—preserving the natural imbalance, diversity, and temporal structure of real dietary behaviour. The dataset contains 6,925 images, 14,349 dish bounding boxes, and metadata (anonymized user ID, date/time, meal context) from 513 users.

To expose practical challenges, we define three benchmark tasks (Figure 4): (i) **closed-set classification** on single-dish crops, (ii) **continual user-aware adaptation** in which a model accumulates user data incrementally over time, and (iii) **context-aware multi-dish classification** using menu-level descriptions. We evaluate ten models spanning CLIP [30], EVA-CLIP [32], and five LMMs using the lmms-eval framework [20, 36]. Key findings include:

- Even GPT-4o achieves only R@1 = 42.4% (F1 = 0.373), demonstrating substantial room for improvement.
- FoodLogAthl-218 exhibits significantly higher intra-class spread and lower cluster compactness than UEC-Food 256.
- Continual LoRA fine-tuning of Qwen2.5-VL-3B raises accuracy from ∼20% to over 50% when using all-user data; user-specific LoRA increases accuracy by ∼50 percentage points for a heavy logger.
- Adding menu-level context raises F1 from 0.159 to 0.230 and P@1 from 0.102 to 0.179, but at a cost to recall.

Our contributions are: (1) a publicly released, ethics-vetted real-world food dataset with rich metadata; (2) a reproducible mixed automated–human curation pipeline; (3) three realistic benchmark tasks with comprehensive baselines.

2 Related Works

Food image datasets. Existing food image benchmarks differ significantly in scale, cuisine coverage, and collection method (Table 2). PFID [7] targets fast-food chains with structured imagery. Food-101 [6] and UPMC Food-101 [34] are large web-crawled collections covering Western cuisine. UEC-Food 256 [18] broadens to 256 classes including Japanese dishes. FoodAI-756 [31] and Food2K [27] scale to hundreds of thousands of web images. UNICT-FD1200 [10] collects images from experiment participants, approaching real-world conditions, but remains small. Food-500 Cap [24] adds fine-grained captions but still relies on web sources. FoodLogAthl-218 is the first publicly available dataset sourced directly from a dietary management app, capturing authentic logging noise, long-tail class distributions, and temporal user behaviour.

Food recognition methods. Early approaches employed CNNs [13, 17] trained on fixed class sets. Personalized classifiers [14, 19, 35] demonstrated that adapting to individual users substantially improves accuracy in real logging conditions. More recently, large vision–language models (LMMs) such as GPT-4o [1], LLaVA [21, 22], and Qwen2-VL [5] have shown strong zero-shot visual recognition capabilities. CLIP-based models [11, 30, 32] enable zero-shot classification through image–text cosine similarity. FoodLogAthl-218 benchmarks all of these paradigms under a unified, realistic evaluation protocol.

Continual and personalized learning. Continual learning methods aim to accumulate knowledge from a stream of data without catastrophic forgetting [29]. Parameter-efficient fine-tuning via

LoRA [16] has emerged as a practical strategy for adapting large models with limited data. Personalized food recognition using individual logs [2, 14, 19, 35] shows that user-specific adaptation is particularly beneficial in the food domain. Our Task 2 benchmark directly evaluates these capabilities on temporally ordered, real user data.

3 FoodLogAthl-218 Dataset

3.1 Data Source and Initial Collection

FoodLog Athl [28] is a dietary management platform that allows users to photograph meals and receive automatic nutritional feedback. Users photograph their meals, the system proposes dish identifications from a 140K-item food database, and users can accept or correct the labels. Each record includes: an anonymized user ID, date/time of the meal, dish names (as corrected by the user), portion sizes, and bounding boxes produced by the detection model.

We collected records from May 2023 to October 2024, yielding an initial corpus of 11,460 images, 61,414 bounding boxes, and 5,889 raw class labels from 632 users. All image collection was conducted under ethics board approval with informed consent from participating users. Two-stage manual PII removal was applied prior to any analysis.

3.2 Data Curation Pipeline

Raw app data contains many artefacts: detection failures, label noise, visually ambiguous or rare classes, and accidental PII exposure. We apply six sequential filtering steps, detailed in Table 1.

(1) **Remove records without images**: images missing from storage are dropped, reducing bounding boxes from 61,414 to 26,646 and classes from 5,889 to 3,167.
(2) **Remove records without bounding boxes**: entries lacking valid bounding box annotations are discarded (11,285 images, 26,065 boxes, 3,071 classes).
(3) **Drop rare classes**: classes with fewer than 10 samples are removed to ensure statistical reliability, leaving 9,121 images, 21,944 boxes, and 1,180 classes.
(4) **CLIP-based outlier filtering**: for each class we compute a centroid in CLIP ViT-B/16 feature space; samples whose feature distance from the centroid exceeds 1.5 standard deviations are flagged and removed, yielding 7,537 images and 17,513 boxes.
(5) **Manual review**: human annotators correct mislabelled bounding boxes, remove invalid annotations, and re-examine flagged borderline cases. To preserve multi-dish meal composition, images are accepted or rejected *as atomic units*—removing a single erroneous dish forces removal of the whole meal photo only when no correction is feasible. This step produces the final 6,925 images, 14,349 boxes, and 513 users.
(6) **Label consolidation**: GPT-4o is prompted to group synonym labels (e.g. regional spelling variants, brand names referring to the same dish) into semantic clusters; annotators then manually refine these clusters and merge them into 218 final classes.

3.3 Dataset Characteristics

The 218-class dataset exhibits a pronounced long-tail distribution (Figure 2), with a few popular dishes (e.g. white rice, miso soup) dominating and many dishes appearing infrequently. Multi-dish images are common, reflecting real Japanese meal customs. Sample images illustrating lighting variation, plating diversity, and multi-dish composition are shown in Figure 1.

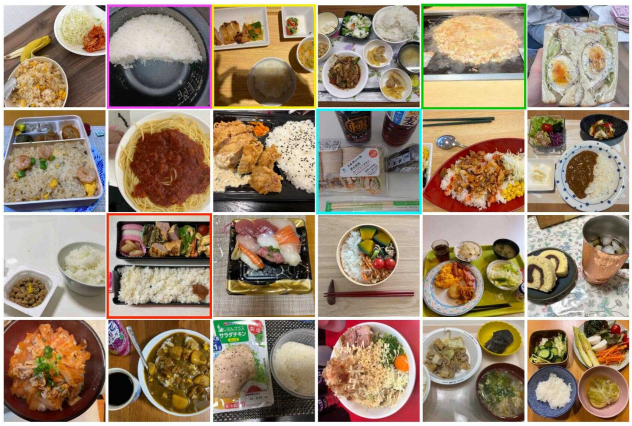

Figure 1: Examples of images in FoodLogAthl-218. Photos are captured in real logging conditions: varying angles, lighting, and plating styles.

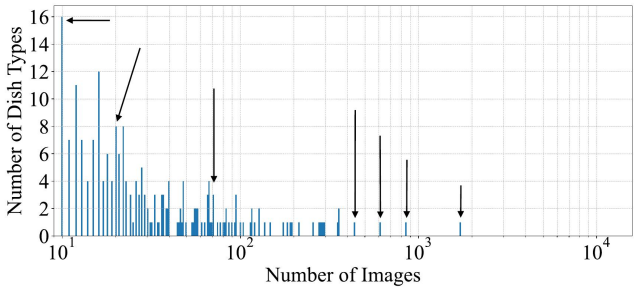

Figure 2: Log–linear distribution of samples per class in FoodLogAthl-218, illustrating the natural long-tail structure. Example dish images are overlaid for selected classes.

To quantify dataset difficulty relative to an established benchmark, we compute four distribution metrics—Trace (scatter), Silhouette, Davies–Bouldin, and Calinski–Harabasz—on features extracted by CLIP ViT-L/14, EVA-CLIP (ViT-g), and Swin Transformer [23], comparing against UEC-Food 256 [18] (Table 3). FoodLogAthl-218 achieves higher Trace and Davies–Bouldin (greater feature spread and inter-class separation ambiguity) and lower Silhouette and Calinski–Harabasz (less compact, more overlapping clusters) across all feature extractors, confirming that it is a more challenging benchmark than UEC-Food 256 (Figure 3).

4 Benchmark Tasks and Experimental Setup

4.1 Task Overview

We define three complementary tasks (Figure 4) that together cover the key challenges of real-world food recognition:

Task 1 — Closed-set classification. Given a bounding-box crop of a single dish, predict its class from the 218-class vocabulary.

Table 1: Data Filtering Process and the Number of Images, Bounding Boxes (bboxes) and Classes.

Step	Operation	#images	(diff)	#bboxes	(diff)	#classes	(diff)
0	Raw data	11,460		61,414		5,889	
1	Remove samples w/o image	11,460	(0)	26,646	(-34,768)	3,167	(-2,722)
2	Remove samples w/o bounding box	11,285	(-175)	26,065	(-581)	3,071	(-96)
3	Remove classes w/ <10 samples	9,121	(-2,164)	21,944	(-4,121)	1,180	(-1,891)
4	Remove outliers based on image features	7,537	(-1,584)	17,513	(-4,431)	1,180	(0)
5	Manual review and remove invalid images	6,925	(-612)	14,349	(-3,164)	1,180	(0)
6	Merge similar classes and clean labels	6,925	(0)	14,349	(0)	218	(-982)

Table 2: Comparison of Datasets for Food Image Classification Tasks.

Year	Dataset Name	#classes	#samples	Cuisine Category	Data Source
2009	PFID [7]	101	4,545	Western	Fast Food Chains
2014	UEC-Food 256 [18]	256	25,088	International	**Web** + Original Database
2014	Food-101 [6]	101	101,000	Western	**Web** (foodspotting.com)
2015	UPMC Food-101 [34]	101	90,840	Western	**Web** (Google [12])
2016	UNICT-FD1200 [10]	1,200	4,754	International	Experiment Participants
2017	Vegfru [15]	25/292*	160,000	Vegetables, Fruits	**Web** (Google, ImageNet [8], etc.)
2019	FoodAI-756 [31]	152/756*	400,000	International	**Web** (Google, Bing, etc.)
2021	Food2K [27]	2,000	1,036,564	International	**Web** (Meituan [25])
2023	Food-500 Cap [24]	494	24,700	International	**Web** (Google, Bing)
2025	FoodLogAthl-218	218	14,349	International	**Dietary Management App**

* Hierarchically structured dataset, where n/m indicates m classes divided into n higher-level classes.

Table 3: Comparison of distribution metrics between FoodLogAthl-218 and UEC-Food 256 using feature extractors. Higher *Trace* and *Davies–Bouldin* indicate greater spread, while lower *Silhouette* and *Calinski–Harabasz* indicate less compact clusters.

	Trace (↑)			Silhouette (↓)			Davies–Bouldin (↑)			Calinski–Harabasz (↓)		
Dataset	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**
UEC-Food 256	71.21	45.09	264.00	0.0335	0.0577	-0.0064	3.64	3.02	4.11	74.05	115.80	49.17
FoodLogAthl-218 (Ours)	**81.70**	**49.92**	**282.57**	**-0.0210**	**0.0036**	**-0.0288**	**4.52**	**3.68**	**4.51**	**27.51**	**51.51**	**21.29**

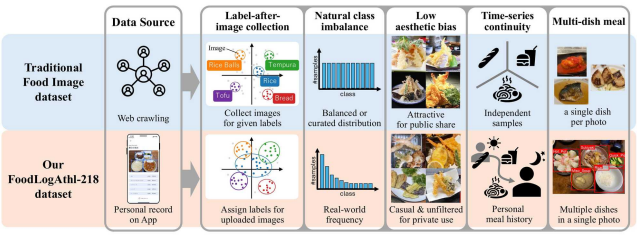

Figure 3: Comparison of feature distributions between traditional food datasets and FoodLogAthl-218. FoodLogAthl-218 shows higher intra-class spread and inter-class overlap, reflecting real-world diversity.

This mirrors the standard food recognition setup and provides a baseline for all models.

Task 2 — Continual user-aware adaptation. Data is presented in chronological order, simulating a user who accumulates meal logs over time. At each weekly checkpoint a model is evaluated on that week's unseen data after potentially being updated on all prior data. Performance is reported as weekly top-1 accuracy together with a weighted one-month moving average (±2 weeks), where weights are proportional to weekly data volume. This task evaluates the benefit of continual learning and user personalization.

Task 3 — Context-aware multi-dish classification. When multiple dishes appear in a single image, the whole-image context (menu-level description) may disambiguate individual dishes. Given a complete meal image with per-dish bounding boxes and a natural-language description of the meal, predict the class of each dish. This task examines whether context improves or introduces noise into per-dish predictions.

4.2 Evaluation Metrics

For Tasks 1 and 3 we report macro-averaged F1, Precision@1 (P@1), and Recall@K for K ∈ {1,3,5}. For Task 2 we report top-1 accuracy per week and the weighted moving average described above.

4.3 LMM Inference Protocol

LMMs generate free-form dish descriptions in response to a visual prompt. To map these to a fixed class label we embed both the generated description and all 218 class-name strings using the `intfloat/e5-large-v2` sentence encoder [33] and assign the class

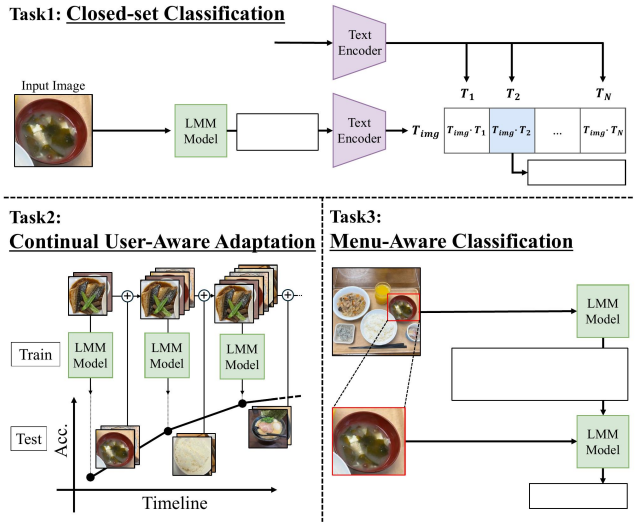

Figure 4: Overview of the three benchmark tasks defined in FoodLogAthl-218. Task 1: closed-set classification on dish crops. Task 2: continual user-aware adaptation. Task 3: context-aware multi-dish classification.

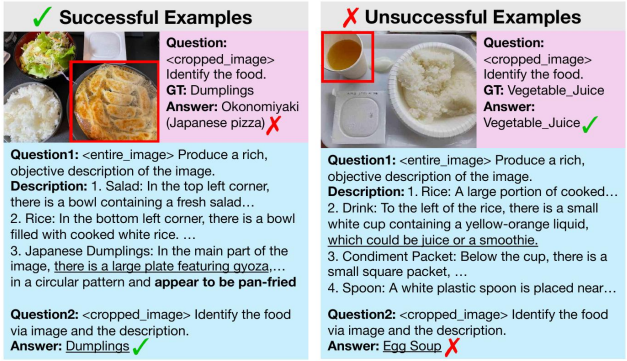

Figure 5: Examples of the description-guided classification task (Task 3). A whole-meal description is generated from the full image and used as additional context for per-dish classification.

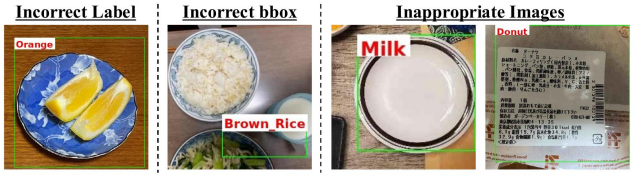

Figure 6: Examples of erroneous model predictions. Common failure modes include visually similar dishes (inter-class confusion), non-standard viewpoints, and heavily processed/mixed foods.

whose embedding has the highest cosine similarity to the generated description. This embedding-based classification allows LMMs to participate in the closed-set evaluation without requiring class-constrained decoding.

4.4 Models

We evaluate the following models:

- **GPT-4o** [1]: closed-source frontier LMM.
- **Qwen2.5-VL-7B-Instruct** and **Qwen2-VL-7B/3B-Instruct** [5]: open-weight LMMs.
- **LLaVA-OneVision-Qwen2-7B** [21]: open-weight LMM based on Qwen2.
- **LLaVA-v1.6-34B** and **LLaVA-v1.6-Mistral-7B** [22]: LLaVA series.
- **LLaVA-v1.5-7B** [22]: earlier LLaVA baseline.
- **EVA-CLIP (ViT-g)** [32] and **CLIP (ViT-L/14)** [30]: contrastive models evaluated by computing cosine similarity between image features and class-name text embeddings.

For Task 2 we additionally fine-tune **Qwen2.5-VL-3B** using LoRA [16] in two settings: (a) fine-tuning on all accumulated user data (*all-users LoRA*) and (b) fine-tuning exclusively on a target user's data (*user-specific LoRA*).

4.5 Task 3: Description-Guided Setup

For Task 3 we use the whole-meal image as additional context. As illustrated in Figure 5, a natural-language description of the full meal is generated by an LMM and provided alongside each dish crop at inference time. We evaluate Qwen2.5-VL-7B-Instruct with and without this menu description.

5 Results

5.1 Task 1: Closed-Set Classification

Table 4 reports closed-set classification performance for all models. GPT-4o achieves the best F1 (0.373) and R@1 (0.424) among LMMs, while CLIP ViT-L/14 obtains the highest R@5 (0.600) despite a lower R@1. This highlights that CLIP's embedding-based similarity excels at shortlisting candidates but struggles with the final selection. All models perform substantially below practical utility thresholds, confirming that FoodLogAthl-218 poses a genuinely difficult benchmark. Smaller open-weight LMMs, such as LLaVA-v1.5-7B, fall below 10% R@1, confirming that general-purpose visual instruction tuning alone is insufficient for fine-grained food recognition. The embedding-based label assignment protocol can introduce additional noise when class names are semantically similar; future work should explore constrained decoding or list-conditioned generation to mitigate this. Figure 6 shows representative error cases: visually similar dishes, unusual viewpoints, and heavily processed foods account for a large proportion of mistakes.

5.2 Task 2: Continual User-Aware Adaptation

Figure 7 shows incremental top-1 accuracy over 35 weeks for zero-shot inference, full fine-tuning, and LoRA fine-tuning on all-user data for Qwen2.5-VL-3B. Zero-shot accuracy remains near 20% throughout, while LoRA fine-tuning steadily improves to exceed 50%, a gain of more than 30 percentage points. Full fine-tuning shows comparable or slightly lower performance than LoRA in

Table 4: Classification performance on FoodLogAthl-218.

Model	F1	R@1	R@3	R@5
GPT-4o	**0.3732**	**0.4243**	**0.4991**	0.5221
Qwen2.5-VL-7B-Instruct	0.2349	0.3684	0.4609	0.4859
Qwen2-VL-7B-Instruct	0.1594	0.3656	0.4590	0.4781
Qwen2-VL-3B-Instruct	0.1386	0.2663	0.3250	0.3517
llava-onevision-qwen2-7b-ov	0.1546	0.3629	0.4497	0.4639
EVA-CLIP (ViT-g)	0.2465	0.2654	0.4897	0.5944
CLIP (ViT-L/14)	0.2393	0.2567	0.4885	**0.6000**
llava-v1.6-34b	0.1354	0.2432	0.3779	0.4063
llava-v1.5-7b	0.0991	0.2140	0.2721	0.2973
llava-v1.6-mistral-7b	0.0944	0.1932	0.2595	0.2831

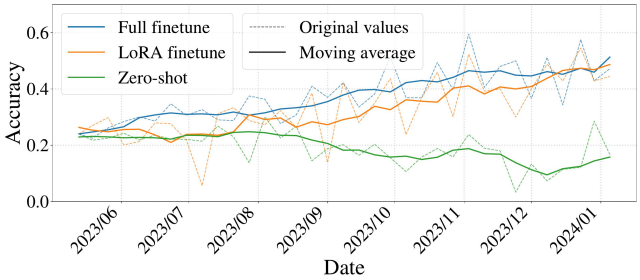

Figure 7: Incremental top-1 accuracy over 35 weeks for zero-shot, full fine-tuning, and LoRA fine-tuning of Qwen2.5-VL-3B on all-user data. Solid lines show per-week accuracy; shaded bands show the weighted one-month moving average.

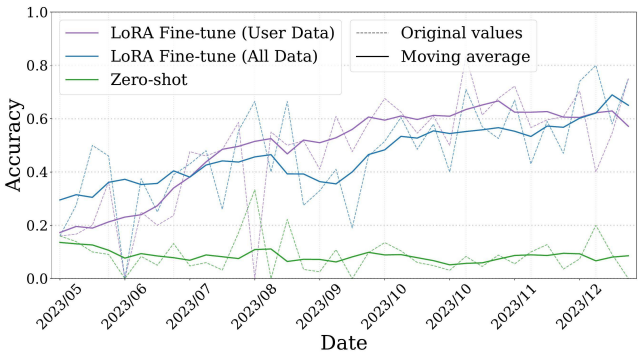

Figure 8: Top-1 accuracy for User A over time: zero-shot, all-users LoRA, and user-specific LoRA. User-specific adaptation substantially outperforms the global model.

early weeks but converges similarly, confirming that parameter-efficient adaptation is sufficient.

Figure 8 zooms into a heavy-logging user (User A). The user-specific LoRA (trained solely on User A's historical data) eventually outperforms the all-users LoRA by approximately 50 percentage points, demonstrating the substantial benefit of personalization. This finding aligns with prior work on personalized food classifiers [14, 19, 35] and motivates the inclusion of this task as a primary benchmark.

Table 5: F1 score, Precision@1 (P@1), and Recall@K (R@1, R@3, R@5) for Qwen2.5-VL-7B-Instruct on the description-guided classification task.

	F1	P@1	R@1	R@3	R@5
w/o description	0.1594	0.1019	**0.3656**	**0.4590**	**0.4781**
w/ description	**0.2304**	**0.1788**	0.3240	0.4032	0.4336

5.3 Task 3: Context-Aware Multi-Dish Classification

Table 5 reports performance for Qwen2.5-VL-7B-Instruct with and without the menu description. Adding the description raises F1 from 0.159 to 0.230 and P@1 from 0.102 to 0.179, but reduces R@1 from 0.366 to 0.324. This precision–recall trade-off arises because menu context sharpens confident predictions while causing the model to over-commit on ambiguous dishes. In aggregate, the description corrects 687 previously wrong predictions but introduces 1,222 new errors, indicating that naive concatenation of context is not always beneficial. More sophisticated context fusion strategies remain an open research problem.

6 Conclusion

We introduced FoodLogAthl-218, a real-world food image benchmark derived from a dietary management application. The dataset captures authentic logging conditions—natural class imbalance, casual photography, multi-dish meal compositions, and temporal user behaviour—that are absent from existing web-sourced benchmarks. A reproducible six-stage curation pipeline combining CLIP-based outlier detection, GPT-4o label grouping, and manual adjudication produces a clean, ethics-vetted resource.

Our evaluation demonstrates that current state-of-the-art models fall far short of practical accuracy on this benchmark. Continual LoRA fine-tuning and user-specific personalization are the most effective remedies identified, suggesting that future systems should incorporate both mechanisms. The context-aware task reveals that menu-level information is a double-edged sword, calling for more principled context integration.

Several limitations of FoodLogAthl-218 warrant acknowledgment. The dataset is confined to Japanese cuisine consumed by competitive athletes, which may limit generalizability to broader dietary populations and global food categories. Bounding boxes produced by the automated detector may retain residual localization errors despite manual review. The embedding-based class-assignment protocol used for LMM evaluation introduces a secondary source of error that is conflated with recognition accuracy. Addressing cross-cuisine generalization, richer nutritional annotations, improved localization quality, and more principled multi-dish context fusion are all promising directions for future work.

FoodLogAthl-218 is released publicly to facilitate research on continual, personalized, and context-aware food recognition in real-world settings.

7 Acknowledgements

We thank all FoodLog Athl users who consented to contribute their meal records to this research. We also thank the annotators who

performed manual review and label consolidation. This work was conducted under ethics board approval with full informed consent.

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